

LEARNING RESOURCE GUIDE

Generative AI for Beginners | Professional Participant Workbook

Section	Section 1: Introduction to Generative AI
Lecture	Lecture 1.2: What is Artificial Intelligence?

Learning Objectives

By the end of this lecture and its accompanying guide, you will be able to:

- Define Artificial Intelligence and distinguish it from rule-based systems.
- Explain the nested relationship between AI, Machine Learning, and Deep Learning.
- Describe the key historical periods in AI development, including the causes of AI winters.
- Identify the three enabling factors that produced the deep learning renaissance in the 2010s.
- Articulate why understanding this history helps professionals evaluate current AI capabilities more accurately.

Introduction

Artificial Intelligence is one of the most used — and most misused — terms in modern discourse. It appears in product marketing, policy debates, news headlines, and boardroom discussions with a breadth that often obscures more than it reveals. Before engaging seriously with generative AI, it is essential to establish a precise understanding of what AI actually is, how it relates to machine learning and deep learning, and where it comes from.

The nested relationship between AI, Machine Learning, and Deep Learning is not merely academic — it determines how different systems behave, what they can and cannot do, and how they fail. A rule-based system fails differently from a machine learning system, which fails differently from a deep neural network. Knowing which type of system you are working with is the first step to working with it responsibly.

The history of AI is also instructive because the field has experienced dramatic cycles of enthusiasm and disappointment. Understanding why these cycles occurred — and what changed to produce the current wave of capability — gives professionals the context to evaluate current claims about AI more critically. Not every breakthrough announced today is as significant as it appears; not every limitation is permanent.

Core Concepts Explained

Artificial Intelligence

Definition: AI is the broad discipline of creating computer systems capable of performing tasks that would typically require human intelligence — including understanding language, recognising images, making decisions, and learning from experience.

Business relevance: The term AI in business contexts often refers to a wide range of technologies with very different behaviours and requirements. Precision about what kind of AI is being deployed enables better procurement, governance, and risk management decisions.

Practitioner relevance: Professionals engaged in AI adoption or governance need to distinguish between narrow AI — systems designed for specific tasks — and broader concepts, to avoid both overestimating capability and underestimating scope.

Real-world example: A thermostat that turns on heating when temperature falls below a set threshold is technically a rudimentary form of AI — it makes a decision based on a rule. ChatGPT is also AI, but the mechanisms involved are vastly more complex. Both are legitimately called AI, which is why the term alone provides little information.

Risks or limitations: Over-relying on the label AI without specifying the type and architecture creates confusion about capability, cost, and risk — leading to misaligned expectations and poor investment decisions.

Machine Learning

Definition: Machine Learning is a subset of AI in which systems learn patterns from data rather than following explicitly programmed rules. The system discovers relationships in data and improves its performance through exposure to examples.

Business relevance: ML enables systems to handle tasks too complex or variable to encode as explicit rules — such as detecting fraud, translating languages, or predicting customer churn. Performance improves as more and better data becomes available.

Practitioner relevance: Understanding that ML systems learn from data — rather than executing fixed logic — helps professionals anticipate how these systems can go wrong: if training data is biased, incomplete, or outdated, the system behaviour will reflect those flaws.

Real-world example: An email spam filter that was explicitly programmed with rules is rule-based AI. A modern spam filter trained on millions of labelled examples is machine learning — it can identify new spam patterns it was never explicitly told about.

Risks or limitations: ML systems can learn spurious correlations from training data. A loan approval model trained on historical data may perpetuate past discriminatory patterns without explicit intent.

Deep Learning

Definition: Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers — hence deep — to learn hierarchical representations of data. It is the technology underpinning virtually all modern generative AI.

Business relevance: Deep learning produced the step-change in capability that powers modern image recognition, language understanding, and content generation. Its effectiveness at scale is what separates current AI tools from earlier generations.

Practitioner relevance: Deep learning models require large amounts of data and significant computational resources to train — factors that shape the economics and accessibility of AI development, which is important context when evaluating claims about deploying custom models.

Real-world example: Image recognition systems that can identify thousands of object categories with human-level accuracy use deep learning. The image generators in tools like Midjourney and DALL-E are also deep learning systems, specifically diffusion models built on deep neural network foundations.

Risks or limitations: Deep learning models are often described as black boxes — it can be difficult to explain precisely why they produced a particular output, which creates challenges for auditing, debugging, and regulatory compliance.

AI Winters and the Deep Learning Renaissance

Definition: AI winters are periods in AI history characterised by reduced funding and interest following failed promises. The deep learning renaissance of the 2010s was driven by the convergence of three factors: massive datasets from the internet, GPU computing power, and algorithmic breakthroughs.

Business relevance: Understanding AI winters contextualises current enthusiasm about the technology. The current wave of capability is backed by genuine technical advances — but so was each previous wave before it stalled. Rigorous evaluation of specific capabilities is more reliable than trend-following.

Practitioner relevance: The three enabling factors — data, compute, algorithms — help explain both what is possible now and what remains constrained. Professionals evaluating AI investments can use this framework to assess whether a proposed application has realistic data and compute prerequisites.

Real-world example: In 2012, a deep learning model outperformed all other approaches in the ImageNet image recognition competition by a significant margin. This single result convinced researchers and investors that deep learning was categorically different — and the modern AI era began.

Practical Applications

Understanding the AI, ML, DL hierarchy and historical context has direct practical value across a range of professional scenarios.

Technology Vendor Evaluation

When evaluating an AI product or vendor, asking what type of AI this is and what learning mechanism it uses provides far more signal than asking simply whether it is AI. A rule-based chatbot and a large language model are both AI, but they have radically different capabilities, maintenance requirements, and failure modes.

AI Risk and Governance Frameworks

Regulatory frameworks for AI, such as the EU AI Act, distinguish between different categories and risk levels of AI systems. Professionals working in compliance or risk functions need to understand these distinctions to apply governance frameworks correctly.

Setting Realistic Expectations with Leadership

Leaders who understand that ML systems learn from data — not from rules — are less likely to expect instant deployment of AI without data infrastructure. This understanding prevents over-promising and under-scoping in AI projects.

Best Practices

- Always specify which type of AI you mean. When discussing AI in professional contexts, be explicit about whether you mean rule-based systems, machine learning, or deep learning. The distinction carries significant implications for capability, cost, and risk.

- Evaluate AI claims against the three enabling factors. When a new AI capability is announced, ask: Does it have sufficient training data? Sufficient compute? Is it built on a proven algorithmic advance? Absence of any of the three suggests the claim may be premature.
- Maintain historical perspective. Current AI capabilities are impressive but not unprecedented in having generated overconfidence. Calibrate enthusiasm with awareness of past hype cycles.
- Connect AI type to failure mode. Rule-based systems fail when rules are incomplete. ML systems fail when data is biased or insufficient. Deep learning systems can fail in unexpected ways due to distribution shift. Know the type; know the failure mode.

Common Mistakes and Misconceptions

Misconception: Using AI, machine learning, and deep learning interchangeably.

Correct approach: These are nested categories with meaningfully different properties. Deep learning is a type of machine learning, which is a type of AI. Not all AI is machine learning; not all machine learning is deep learning.

Misconception: Assuming current AI enthusiasm is just another hype cycle that will fade like previous ones.

Correct approach: While hype cycles have occurred before, the current wave is grounded in genuine, measurable capability advances. The appropriate response is calibrated evaluation, not dismissal.

Misconception: Believing that AI systems are neutral because they learn from data rather than following human-programmed rules.

Correct approach: ML systems reflect the patterns in their training data, including biases, gaps, and historical inequities. Data is not neutral; neither are systems trained on it.

Reference Table: AI, Machine Learning, and Deep Learning — Key Distinctions

Use this table to quickly distinguish between the three nested categories of AI technology when evaluating any AI product or claim.

Dimension	AI (Broad)	Machine Learning	Deep Learning
Definition	Computer systems performing human-like tasks	Learning patterns from data	Multi-layer neural networks learning hierarchical features
Learning mechanism	Rules or learning	Learning from labelled or unlabelled data	Gradient-based optimisation through layers
Data requirement	Low to high	Medium to high	Very high
Compute requirement	Low to medium	Medium	High to very high
Explainability	High (rule-based)	Medium	Often low — black box
Example	Thermostat, chess engine	Spam filter, recommendation engine	ChatGPT, DALL-E, speech recognition
Main failure mode	Incomplete rules	Data bias or insufficient data	Distribution shift, hallucination, opacity

How to use this: When encountering a new AI product or claim, use this table to identify which category it belongs to — then apply the relevant considerations for that category.

Knowledge Check

Test your understanding before moving on. Answers are shown below each question.

1. A company is evaluating an AI-powered customer service system. The vendor says it uses AI. What follow-up question would be most useful for the evaluation team?

- A. How many employees does the vendor have?
- B. Is this a rule-based system, a machine learning model, or a deep learning model — and what training data was used?
- C. What year was the company founded?
- D. Does the vendor offer 24/7 support?

Answer: B. Is this a rule-based system, a machine learning model, or a deep learning model — and what training data was used?

Why: *The type of AI determines capabilities, failure modes, data dependencies, and governance requirements — all essential for informed evaluation.*

2. Which of the following best explains why AI winters occurred in the past?

- A. Researchers lost interest in the field.
- B. Government regulations shut down AI research.
- C. The technology failed to deliver on promised capabilities, often due to insufficient data and compute.
- D. All important AI problems were already solved.

Answer: C. The technology failed to deliver on promised capabilities, often due to insufficient data and compute.

Why: *AI winters resulted from the gap between promise and technical reality — specifically the lack of the enabling factors that made the 2010s different.*

3. A compliance officer reviewing an AI credit risk system notes lower approval rates for a specific demographic group. Which property of machine learning best explains this?

- A. ML systems are programmed with explicit discriminatory rules.
- B. ML systems learn from historical data and can perpetuate biases present in that data.
- C. Deep learning systems are always more accurate than rule-based systems.
- D. Credit risk models are not considered AI.

Answer: B. ML systems learn from historical data and can perpetuate biases present in that data.

Why: *Machine learning systems reflect patterns in training data, including historical inequities. Biased inputs produce biased outputs, even without explicit intent.*

Reflection Questions

1. Before this lecture, how did you define Artificial Intelligence? How has your understanding changed, and what distinction is most useful for your professional context?
2. Can you identify an AI system you interact with regularly and classify it — is it likely rule-based, machine learning, or deep learning? What evidence supports your classification?

3. The three enabling factors for the current AI wave were data, compute, and algorithms. Which of these do you think is the most significant constraint for AI development in your industry?
4. AI winters followed periods of overconfidence. How would you apply this historical pattern to evaluate AI claims you encounter today in your professional environment?
5. If machine learning systems reflect their training data, what questions should you ask about the training data of any AI system you are considering adopting?

Key Takeaways

- AI is a broad category; Machine Learning and Deep Learning are progressively more specific subsets with distinct mechanisms, requirements, and failure modes.
- Not all AI learns — rule-based systems execute explicit logic without learning from data.
- The deep learning renaissance was driven by three converging enabling factors: large datasets, GPU computing power, and algorithmic breakthroughs.
- AI winters resulted from genuine gaps between promised and actual capability — a pattern worth keeping in mind when evaluating current AI claims.
- ML systems reflect patterns in their training data, including biases — data quality is inseparable from AI system quality.
- Deep learning models are often difficult to interpret, creating governance challenges that require explicit attention in regulated environments.
- Understanding the AI, ML, DL hierarchy enables more precise professional conversations about capability, risk, and governance.

Recommended Next Actions

6. Draw the nested circle diagram — AI contains ML, ML contains Deep Learning — from memory and annotate each circle with definition, data requirement, and one example system.
7. Identify one AI claim you have heard recently from a vendor, colleague, or news article and apply the AI, ML, DL framework to categorise it more precisely.
8. Look up one AI system used in your industry and try to determine whether it is rule-based, ML-based, or deep learning-based. What evidence can you find?
9. Add to your glossary: Artificial Intelligence, Machine Learning, Deep Learning, AI Winter, GPU.